Mathematical Foundations of Computational Electrophysiology

Gap-Analysis Revision: Eleven New Chapters

CON^{2}PHYS mathematical companion

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Chapter 1

The measure-theoretic vocabulary behind probability

A probability model is not merely a formula for averages. It states *which outcomes exist*, *which events can be assigned probabilities*, and *which calculations are allowed*. This chapter provides the missing hypotheses for Results 21–24, 40–46, and 71–76.

Definition 1.1 (Sigma-algebra and measure)**.** *For a set Ω, a σ-algebra $\mathcal{F}$ is a family of subsets containing Ω, closed under complements and countable unions. A measure $\mu : \mathcal{F} \to [0, \infty]$ satisfies $\mu(\varnothing) = 0$ and countable additivity over disjoint events. A probability measure has $\mu(\Omega) = 1$. A measure is* σ-finite *if Ω is a countable union of measurable sets of finite measure.*

Definition 1.2 (Random variable and expectation)**.** *A real-valued random variable X is a measurable map $(\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$: for every Borel set B, its preimage $X^{-1}(B)$ is an event. If $\int |X| \, d\mathbb{P} < \infty$, define $\mathbb{E}X = \int X \, d\mathbb{P}$. The space L^2 consists of equivalence classes of variables with $\mathbb{E}|X|^2 < \infty$; equality almost surely is identified. Its inner product is $\langle X, Y \rangle = \mathbb{E}[X\overline{Y}]$. A* Hilbert space *is a complete inner-product space; L^2 is one. Completeness is what makes orthogonal projection possible.*

Proposition 1.3 (Conditional expectation is a projection)**.** *For $X \in L^2$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is characterized by being $\mathcal{G}$-measurable and satisfying $\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])Z] = 0$ for all bounded $\mathcal{G}$-measurable Z. It is the orthogonal projection of X onto the closed subspace $L^2(\mathcal{G})$.*

Proof. The defining conditional-expectation identity gives orthogonality for bounded Z. Truncation followed by L^2 approximation extends it to $Z \in L^2(\mathcal{G})$. The projection theorem then characterizes the minimizer of $\mathbb{E}[(X - Z)^2]$ among such Z. □

This explains why a conditional firing-rate prediction is an optimal mean-square prediction *relative to the variables conditioned upon*, not relative to every unobserved brain state.

Definition 1.4 (Stationarity, ergodicity, and predictability)**.** *A process X_t is* wide-sense stationary *if its mean is constant and $\mathrm{Cov}(X_t, X_s)$ depends only on $t - s$. It is* strictly stationary *if all finite-dimensional joint laws are time-shift invariant. An invariant process is* ergodic *when invariant events have probability zero or one. A predictable intensity $\lambda(t \mid \mathcal{H}_{t-})$ is measurable relative to information available* just before *t; this forbids predicting a spike using its own occurrence. The time-integrated intensity is called its* compensator *when the Doob–Meyer conditions are satisfied. For a regular parametric model p_θ, the* score *is $U_\theta = \nabla_\theta \log p_\theta(X)$ and* Fisher information *is $\mathcal{I}(\theta) = \mathbb{E}_\theta[U_\theta U_\theta^\top]$, under assumptions justifying its finiteness and calculus.*

Proposition 1.5 (Law of total variance revisited)**.** *For $X \in L^2$,*

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid \mathcal{G})] + \mathrm{Var}(\mathbb{E}[X \mid \mathcal{G}]).$$

Proof. Write $X - \mathbb{E}X = (X - M) + (M - \mathbb{E}X)$ where $M = \mathbb{E}[X \mid \mathcal{G}]$. The cross term vanishes by orthogonality. Taking squared expectations yields the identity. □

Example 1.6 (LFP recording versus ensemble)**.** If a stationary and ergodic model describes a long LFP trace, its time average can estimate the ensemble mean under integrability. Stationarity alone does not justify equating those means, nor does increasing the sample rate create independent trials.

Chapter 2

A working toolkit for Gaussian models

Gaussian conditioning silently appears in Kalman filters, Gaussian processes, factor analysis, and several multivariate estimators. It belongs before relying on their formulas.

Definition 2.1 (Multivariate normal vector)**.** *A vector $X \in \mathbb{R}^d$ is jointly Gaussian if every scalar linear combination $a^\top X$ is normally distributed (possibly degenerate). Its mean μ and covariance $\Sigma \succeq 0$ characterize its distribution. When $\Sigma \succ 0$, its density is*

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp\left[-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right].$$

Theorem 2.2 (Gaussian conditioning by a Schur complement)**.** *Suppose (X, Y) is jointly normal and partition its mean and covariance into μ_X, μ_Y and blocks Σ_{XX}, Σ_{XY}, Σ_{YY}. If Σ_{YY} is invertible,*

$$\mathbb{E}[X \mid Y = y] = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y),$$
$$\mathrm{Cov}(X \mid Y) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}.$$

Proof. Define $R = X - \mu_X - \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y)$. Then $\mathrm{Cov}(R, Y) = 0$. Joint Gaussianity makes R and Y independent. Expanding $\mathrm{Cov}(R)$ gives the Schur complement. □

Proposition 2.3 (Wishart sample-covariance model)**.** *If $Z_1, \ldots, Z_T$ are independent $\mathcal{N}_d(0, \Sigma)$ with known zero mean, then $\sum_{t=1}^{T} Z_t Z_t^\top$ has a Wishart distribution with scale Σ and T degrees of freedom. The usual centered sample covariance instead involves $T - 1$ degrees of freedom for independent Gaussian samples.*

This distributional assumption is the model behind classical random-covariance results; the same formula does not describe arbitrarily correlated neuronal time bins.

Proposition 2.4 (Multivariate Stein identity)**.** *For $X \sim \mathcal{N}_d(\mu, \Sigma)$ and a differentiable function f with appropriate integrability and boundary decay,*

$$\mathbb{E}[(X - \mu)f(X)] = \Sigma\,\mathbb{E}[\nabla f(X)].$$

Proof. For a nondegenerate Gaussian density p, $\nabla p(x) = -\Sigma^{-1}(x - \mu)p(x)$. Integrate $\nabla(fp)$ and use the vanishing boundary term. □

2.1 Gaussian KL divergence

For full-rank Gaussian distributions $P = \mathcal{N}(\mu_0, \Sigma_0)$ and $Q = \mathcal{N}(\mu_1, \Sigma_1)$ in dimension d,

$$D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}\left[\operatorname{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^\top \Sigma_1^{-1}(\mu_1 - \mu_0) - d + \log\frac{\det\Sigma_1}{\det\Sigma_0}\right].$$

A proof expands $\mathbb{E}_P[\log p(X) - \log q(X)]$, then uses the covariance trace identity. The formula requires positive-definite covariance matrices; singular distributions need measure-theoretic treatment of their supports.

Example 2.5 (Neural latent variables)**.** A Kalman measurement update is repeated conditioning of a joint Gaussian state–observation vector. Its familiar gain matrix is therefore a Schur-complement calculation, not a property of all nonlinear or non-Gaussian observation processes.

Chapter 3

From membrane currents to extracellular electrodes

An extracellular electrode does not measure a neuron's voltage in isolation. It measures a difference of electric potential created by spatially distributed ionic currents, tissue geometry, conductivity, referencing, and acquisition electronics[1, 2].

3.1 Electrochemical driving forces

For an ideal dilute permeant ion with valence $z \neq 0$, equality of electrochemical potentials on the two sides of a membrane gives

$$E_{\text{ion}} = \frac{RT}{zF} \log \frac{c_{\text{out}}}{c_{\text{in}}}.$$

This is the Nernst equilibrium potential under its concentration and temperature assumptions. The Goldman–Hodgkin–Katz formula extends the constant-field model to multiple monovalent ionic species with permeability parameters; it is not equivalent to measuring those permeabilities from a single LFP. In the Hodgkin–Huxley description the sodium current is modeled as $I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})$; gating variables obey voltage-dependent kinetic equations. The passive cable approximation has the schematic form

$$c_m \partial_t V(x,t) = g_a \partial_{xx} V(x,t) - g_m (V - E_L) + i_{\text{ext}}(x,t),$$

with coefficients per unit length and consistent units. Cable and membrane models determine transmembrane sources; they are not by themselves electrode forward models.

3.2 Quasistatic field equation

At electrophysiological frequencies and spatial scales where inductive effects are negligible, write $\mathbf{E} = -\nabla\phi$. Let impressed current $\mathbf{J}_p$ and Ohmic return current $-\boldsymbol{\sigma}\nabla\phi$ sum to a divergence-free total current. Current conservation implies

$$\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = \nabla\cdot\mathbf{J}_p, \quad \text{or} \quad -\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = q, \quad q = -\nabla\cdot\mathbf{J}_p.$$

The tensor $\boldsymbol{\sigma}$ may be anisotropic and spatially inhomogeneous. To solve the equation, the domain, conductivity, boundary conditions and electrode reference must be specified. For an infinite isotropic medium, a point source of strength I located at $\mathbf{r}_0$ gives, away from that point,

$$\phi(\mathbf{r}) = \frac{I}{4\pi\sigma|\mathbf{r} - \mathbf{r}_0|}.$$

Superposition yields a discrete source model $\mathbf{v} = G\mathbf{s}$ when the forward physics is linear, with lead-field entries obtained by integrating the Green kernel over chosen source basis functions. A dipole arises as the leading far-field contribution when net source current cancels; line-source models integrate along a dendrite.

Proposition 3.1 (Uniqueness for fixed Dirichlet conditions)**.** *Let Ω be a bounded connected region and $\boldsymbol{\sigma}$ uniformly positive definite. If two weak solutions of the same elliptic equation have equal Dirichlet boundary values, they agree (almost everywhere).*

Proof. Their difference w satisfies the homogeneous equation and zero boundary values. Multiply by w, integrate by parts, and use the boundary condition: $\int_\Omega \nabla w \cdot \boldsymbol{\sigma} \nabla w \, dx = 0$. Positivity implies $\nabla w = 0$; the boundary forces $w = 0$. For pure Neumann conditions, uniqueness holds only up to an additive constant, with a compatibility condition on source flux. □

3.3 Why the inverse need not be unique

Suppose an electrode array measures $\mathbf{v} = G\mathbf{s}$. If $G\mathbf{h} = 0$ for nonzero $\mathbf{h}$, then $G(\mathbf{s} + \mathbf{h}) = G\mathbf{s}$. This is exact finite-dimensional nonidentifiability. In the continuum, smoothing operators can also be ill-conditioned or compact, so tiny measurement errors may cause enormous source errors even when uniqueness holds in a restricted model. *Hadamard well-posedness* requires existence, uniqueness and continuous dependence of solutions on data. A linear operator is *compact* if it maps bounded sets into sets with compact closure; its singular values in infinite-dimensional inverse problems can converge to zero. For singular system $Gv_j = \sigma_j u_j$ and data b, the formal inverse contains $\langle b, u_j \rangle / \sigma_j$ and magnifies noise when $\sigma_j \to 0$. A classical Picard-type condition controls the squared normalized coefficients for compatible data. Tikhonov regularization instead solves

$$\widehat{\mathbf{s}}_\lambda = \arg\min_{\mathbf{s}} \{ \|G\mathbf{s} - \mathbf{v}\|_2^2 + \lambda \|\mathbf{s}\|_2^2 \} = (G^\top G + \lambda I)^{-1} G^\top \mathbf{v}.$$

Example 3.2 (MATLAB and interpretation)**.** `field_potential_forward.m` receives G; it does *not* compute tissue conductivity or the Green kernel. `field_potential_inverse_tikhonov.m` returns one regularized source estimate, not a unique membrane-current pattern. `current_source_density.m` needs spatial geometry and differentiated potentials; references and volume conduction remain crucial.

Chapter 4

Digital filtering, stability, and aliasing

The first operation in many LFP workflows is band-pass filtering. Its interpretation requires a distinction between a sampled sequence, its discrete-time filter, and the underlying analog field[7].

4.1 Poisson summation and aliasing

For suitable rapidly decreasing $x(t)$, Fourier convention $X(f) = \int x(t)e^{-2\pi ift}\,dt$, and sampling frequency F_s, Poisson summation gives the periodic spectrum of its samples:

$$\sum_{n\in\mathbb{Z}} x(n/F_s)e^{-2\pi inf/F_s} = F_s \sum_{k\in\mathbb{Z}} X(f + kF_s).$$

The observed spectrum is thus a *sum of translated copies*. The copies overlap unless the analog spectrum is sufficiently bandlimited; an anti-alias filter must act *before* digitization. As a concrete counterexample, the sequences from frequencies f and $f + F_s$ agree because $e^{2\pi i(f+F_s)n/F_s} = e^{2\pi ifn/F_s}$. No post-acquisition algorithm can distinguish them without additional prior information.

4.2 Convolution, the z-transform, and poles

For a causal linear time-invariant discrete filter, $y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$, with

$$H(z) = \sum_{k=0}^{\infty} h[k]z^{-k}.$$

Its frequency response is $H(e^{i\omega})$ when that boundary value exists. Absolute summability $\sum_k |h[k]| < \infty$ is sufficient for bounded-input bounded-output (BIBO) stability. For rational causal filters in minimal form, BIBO stability requires all poles strictly inside the unit circle.

Proposition 4.1 (First-order recursive filter)**.** *For $y[n] = ay[n-1] + x[n]$ with zero initial condition, $h[k] = a^k \mathbf{1}_{k\geq 0}$. The system is BIBO stable if and only if $|a| < 1$.*

Proof. The absolute impulse-response sum is $\sum_{k\geq 0} |a|^k = (1 - |a|)^{-1}$ when $|a| < 1$. If $|a| \geq 1$, the impulse response is not absolutely summable; there exists a bounded input producing an unbounded output. □

4.3 FIR versus IIR; delay and zero-phase processing

An FIR filter has a finite impulse response and is BIBO stable. An IIR filter may have sharper transitions for a given order, but its poles and finite-precision implementation need inspection. For nonzero frequency response, define group delay $\tau_g(\omega) = -d \arg H(e^{i\omega})/d\omega$. A forward-backward filter ideally has response $H(e^{i\omega})H(e^{-i\omega}) = |H(e^{i\omega})|^2$ for real filter coefficients. This suppresses phase delay but is *noncausal*, changes the effective magnitude response, and requires boundary extrapolation: edge transients are not removed. An ideal brick-wall filter has an infinitely long sinc-shaped impulse response; abrupt truncation creates Gibbs-type ringing.

Example 4.2 (What the arsenal computes)**.** `bandpass_lfp.m` designs a Butterworth band-pass, converts it to second-order sections and applies `filtfilt`. The returned trace is a zero-phase *offline estimate*, not the output of a causal online instrument. The initial and final samples need explicit transient inspection. Filtering cannot remove volume conduction, spike-sorting errors or physiological common inputs.

Chapter 5

Time–frequency representations and estimator uncertainty

A discrete Fourier transform can be computed exactly while an estimate of an unknown stochastic power spectrum remains noisy. The distinction is central to electrophysiological inference.

5.1 Why the raw periodogram is not generally consistent

For a zero-mean Gaussian white-noise series with variance σ^2 and length T, define at a non-DC Fourier frequency

$$I_T(f_k) = \frac{1}{T}\left|\sum_{t=0}^{T-1} X_t e^{-2\pi i k t/T}\right|^2 .$$

At generic positive Fourier frequencies below Nyquist, real and imaginary Fourier coefficients are independent normals with the appropriate variance; hence $2I_T(f_k)/\sigma^2 \sim \chi^2_2$ under this exact white Gaussian model. Consequently the periodogram variance does not vanish as T increases. Averaging approximately independent tapered or segmented estimates reduces variance, at a cost of frequency resolution and possible bias. For K independent identical exponential spectral ordinates, $2K\bar{I}/S \sim \chi^2_{2K}$; Welch overlap and tapered spectra violate exact independence, so their effective degrees of freedom need calibration.

5.2 Short-time Fourier transform, wavelets and admissibility

A window w produces the short-time Fourier transform $V_w x(\tau, f) = \int x(t)\overline{w(t-\tau)}e^{-2\pi i f t}\,dt$. Window duration sets an uncertainty trade-off; the display resolution is not a ground-truth measurement of a neural oscillation's instantaneous frequency. For a continuous wavelet ψ, the usual admissibility condition is

$$C_\psi = \int_{\mathbb{R}} \frac{|\Psi(f)|^2}{|f|}\,df < \infty.$$

It requires zero mean for sufficiently regular wavelets, so an uncorrected finite-frequency Morlet wavelet is only approximately admissible; a DC correction makes the distinction explicit. Reconstruction follows with the appropriate wavelet normalization. Discrete wavelet frames require bounds $A\|x\|^2 \le \sum_j |\langle x, \psi_j\rangle|^2 \le B\|x\|^2$ for positive finite A, B.

5.3 Coherence, phase consistency and finite samples

Even when the population coherence is zero, the squared magnitude of a finite-sample coherence estimate is nonnegative and generally biased upward. A simple idealized average of K indepen-

dent complex Gaussian spectral replicates gives, under zero population coherence, an estimator with $\mathbb{E}[\widehat{C}] \approx 1/K$; this is *not* a universal threshold for overlapped, tapered or strongly dependent data. For iid spike phases $\phi_1, \ldots, \phi_N$, the usual phase-locking value (PLV) is $|N^{-1} \sum_j e^{i\phi_j}|$. Even uniformly distributed phases produce a positive finite-sample value. The pairwise phase consistency (PPC)

$$\widehat{\mathrm{PPC}} = \frac{1}{N(N-1)} \sum_{j \neq k} \cos(\phi_j - \phi_k)$$

has zero expectation for independent uniform phases, provided the sampling and independence conditions hold. Surrogates made by Fourier phase randomization typically preserve individual Fourier magnitudes; iterative amplitude-adjusted Fourier transforms (IAAFT) approximately preserve both the marginal amplitude distribution and a target power spectrum. Neither automatically preserves cross-channel dependence or experimental trial structure.

Example 5.1 (What is—and is not—implemented)**.** `welch_psd.m`, `multitaper_psd.m`, `wavelet_coherence.m` and `pairwise_phase_consistency.m` produce numerical estimators. A scientific significance statement requires effective degrees of freedom, trial/block design and an appropriate null; it is not guaranteed by the function name.

Chapter 6

Exponential families, GLMs, and constrained optimization

The Poisson model in `poisson_spike_glm.m` can be derived from a probabilistic model, rather than treated as an unexplained numerical recipe.

6.1 Poisson counts as an exponential family

For count $Y \in \{0, 1, \ldots\}$ and positive mean μ,

$$p(y \mid \mu) = \exp\big(y \log \mu - \mu - \log(y!)\big).$$

Its natural parameter is $\eta = \log \mu$. For a row of covariates $x_i^\top$, exposure duration $\Delta_i > 0$, and regression coefficients β, define $\mu_i = \Delta_i \exp(x_i^\top \beta)$. In matrix form with design X,

$$\ell(\beta) = \sum_i [y_i (x_i^\top \beta + \log \Delta_i) - \Delta_i e^{x_i^\top \beta} - \log(y_i!)].$$

Proposition 6.1 (Concavity of the Poisson log likelihood)**.** *For fixed nonnegative counts and strictly positive exposures,*

$$\nabla \ell = X^\top (y - \mu), \qquad \nabla^2 \ell = -X^\top W X, \quad W = \operatorname{diag}(\mu_i) \succ 0.$$

Thus ℓ is concave; if X has full column rank, it is strictly concave at every finite coefficient vector.

Proof. Differentiate the exponential using

$$\frac{\partial \mu_i}{\partial \beta} = \mu_i x_i.$$

For nonzero v, $-v^\top \nabla^2 \ell v = \sum_i \mu_i (x_i^\top v)^2$, positive under full rank. □

Strict concavity is not existence of a finite maximizer. For all-zero counts and an unpenalized intercept, likelihood increases as the intercept tends to $-\infty$.

6.2 IRLS as Fisher scoring

Set $\eta = X\beta + \log \Delta$ and $\mu = e^\eta$. Newton scoring solves, at the current iterate,

$$(X^\top W X)\, \delta = X^\top (y - \mu), \qquad \beta_{\text{new}} = \beta + \delta.$$

This is equivalent to weighted least squares on the working response $z = \eta + (y - \mu)/\mu$, with the exposure offset kept fixed. For the canonical Poisson family, the observed negative Hessian agrees with expected Fisher information. A quadratic ridge penalty adds $-\lambda P\beta$ to the score and λP to the negative Hessian. If the algorithm *clips* η, or lacks line search, the numerical iterations need not monotonically improve the original unclipped likelihood. This is relevant to the present IRLS draft.

6.3 Jensen, the ELBO, and EM monotonicity

For latent variable Z and any density $q(z)$ supported where the joint model is positive, Jensen gives

$$\log p_\theta(y) = \log \int q(z) \frac{p_\theta(y, z)}{q(z)}\, dz \geq \mathcal{L}(q, \theta) := \mathbb{E}_q[\log p_\theta(y, Z)] - \mathbb{E}_q[\log q(Z)].$$

The identity $\log p_\theta(y) = \mathcal{L}(q, \theta) + D_{\mathrm{KL}}(q \| p_\theta(z \mid y))$ shows exactly when the lower bound is tight. Exact EM alternates an E-step setting q to the posterior and an M-step not decreasing the ELBO. This gives nondecreasing observed-data likelihood, but *not* a global optimum. For phase–amplitude coupling, the Tort modulation index is a normalized KL divergence between a binned phase-dependent amplitude distribution and the uniform distribution; positivity comes from Gibbs' inequality, while binning and surrogates are additional statistical decisions.

6.4 KKT and alternating least squares

For differentiable convex f with inequality constraints $g_j(\beta) \leq 0$ and equality constraints $h_k(\beta) = 0$, KKT conditions (with appropriate constraint qualifications) require stationarity, feasibility, nonnegative multipliers and complementary slackness. Sparse Lasso penalties require subgradients rather than ordinary derivatives at zero. For a CP tensor decomposition, exact minimization over a factor block cannot increase the unpenalized objective *of that specified block problem*; alternating such steps does not make the multi-factor objective convex or guarantee global optimality. The exploratory `tensor_cp_als.m` includes ridge terms and does not track a monotonic objective; a monotonicity claim for that code must be tested, not assumed.

Chapter 7

Renewal processes, spike-train spectra, and bin limits

A spike train may be treated as a random counting measure $n(t) = \sum_j \delta(t - t_j)$, where the Dirac deltas are *distributions*, not finite-amplitude voltage spikes. This representation connects point processes to Fourier spectra and to continuous-time LFP shot-noise models.

7.1 How Bernoulli bins approach a Poisson process

Partition a fixed duration T into $m = T/\Delta$ bins, each with independent event probability $p_\Delta = \lambda\Delta$. Then $N_T \sim \text{Binomial}(m, p_\Delta)$, with

$$\mathbb{P}(N_T = 0) = (1 - \lambda T/m)^m \longrightarrow e^{-\lambda T}.$$

The binomial probability generating function likewise converges to $\exp[\lambda T(s-1)]$, proving convergence of the *total count* to a Poisson distribution. Independence of disjoint bins and consistency across intervals give the homogeneous Poisson-process limit. A form of Le Cam's inequality bounds total-variation error in Poisson approximations by a constant multiple of $\sum_i p_i^2$; hence fixed T and $p_i = \lambda\Delta$ make this error vanish as $\Delta \to 0$. This approximation is not a guarantee for refractory or bursty neuronal processes.

7.2 Renewal variability

Suppose interspike intervals $S_1, S_2, \ldots$ are iid, strictly positive, with mean m and finite variance v. Their coefficient of variation is $CV^2 = v/m^2$. Under standard renewal limit conditions, for long observation windows T the spike count satisfies

$$\mathbb{E}[N(T)] \sim T/m, \qquad \text{Var}[N(T)] \sim (v/m^3)T, \qquad \frac{\text{Var}\, N(T)}{\mathbb{E}N(T)} \longrightarrow CV^2.$$

Thus *long-window* count Fano factor and ISI CV^2 are related for a renewal model. Nonrenewal history, trial mixing, latent gain, short windows and nonstationarity can break the empirical approximation.

7.3 Bartlett spectra and shot-noise LFP

For a stationary point process, the Bartlett spectrum is the spectrum of its centered event-counting measure, including the shot-noise contribution. A homogeneous Poisson train of rate λ has flat continuous spectral density $S_{nn}(f) = \lambda$ in a two-sided Hz convention (the DC mean

is treated separately). For a stationary renewal process with inter-event characteristic function $\varphi(f) = \mathbb{E}[e^{-2\pi i f S}]$, the nonzero-frequency continuous spectral density takes the form

$$S_{nn}(f) = \lambda \frac{1 - |\varphi(f)|^2}{|1 - \varphi(f)|^2},$$

under the usual nonlattice and regularity assumptions. This shows how refractoriness or regular spiking produces spectral structure without sinusoidal membrane oscillations.

If an idealized LFP component is a linear superposition of spike-triggered kernels, $y(t) = \int h(t - s)\, dN(s)$, its centered spectrum satisfies $S_{yy}(f) = |H(f)|^2 S_{nn}(f)$; independent measurement noise adds its own spectrum. If both n and y derive from the same events, their cross-spectrum is proportional to $H(f)S_{nn}(f)$ with sign/conjugation depending on the cross-spectrum convention. This is an example of *spike–field coherence without a claim of synaptic causation.*

7.4 Hawkes excitation and stability

The classical branching formulation is described by Hawkes[4]. For a linear Hawkes intensity $\lambda(t) = \mu + \sum_{t_j < t} h(t - t_j)$ with $h \geq 0$, $\mu > 0$ and branching ratio $\nu = \int_0^\infty h(s)\, ds < 1$, a stationary version exists under the standard construction, and its mean rate is $\bar{\lambda} = \mu/(1 - \nu)$. When $\nu \geq 1$, this stationary finite-mean construction fails in the usual nonnegative linear model; the statement does not apply without alteration to inhibitory or nonlinear point-process intensities.

Example 7.1 (Arsenal interpretation)**.** Compare `spike_count_fano.m`, `spike_field_coherence.m` and `spike_field_ppc.m` on Poisson, renewal, and latent-rate simulations. A function operating on binned counts or spike phases does not itself implement a continuous-time Bartlett spectral estimator.

Chapter 8

Population geometry, detectability, and identifiability

A low-dimensional fit is not necessarily a detectable or uniquely identified neural mechanism. This chapter separates three questions: *Is a component present in the data? Can the estimator recover its direction? Does that direction have a unique biological meaning?*

8.1 Beyond the Marchenko–Pastur noise edge

Consider an idealized population covariance $\Sigma = I + \theta vv^\top$, with $\|v\| = 1$, $\theta > 0$, iid Gaussian observations, and $N/T \to \gamma \in (0, \infty)$. In the standard spiked-Wishart model, the top population eigenvalue is $1 + \theta$ and the sample-noise edge is $(1 + \sqrt{\gamma})^2$.

Theorem 8.1 (BBP transition, rank-one covariance case)**.** *When $\theta \le \sqrt{\gamma}$, the leading sample eigenvalue approaches the upper noise edge and the sample top direction does not have asymptotically nonzero alignment with v. When $\theta > \sqrt{\gamma}$, the sample leading eigenvalue converges to $(1 + \theta)(1 + \gamma/\theta)$, strictly above the edge, and asymptotic alignment is nonzero.*

The result belongs to the spiked-covariance phase-transition family[5]. It depends on the iid isotropic sampling model. Temporal dependence, heterogeneous neuron variances, preprocessing, and correlated noise modify its calibration. A detectable statistical spike is not automatically an identified cell assembly or an anatomical network.

8.2 Shrinkage is part of covariance estimation

For sample covariance S of N variables and a target T_0 (often proportional to the identity), a linear shrinkage estimator is $\widehat{\Sigma} = (1 - \alpha)S + \alpha T_0$. The optimal α depends on the loss, assumptions, and estimation method. Ledoit–Wolf-type procedures estimate shrinkage to reduce high-dimensional covariance risk[6]; James–Stein concerns shrinkage of normal means in its classical form. Neither is a license to tune hyperparameters on held-out test trials.

8.3 Nonuniqueness of latent coordinates

In Gaussian factor analysis, $y = \Lambda z + \epsilon$ with $z \sim \mathcal{N}(0, I)$, the rotation $\Lambda' = \Lambda Q$, $z' = Q^\top z$ for orthogonal Q leaves the observable covariance $\Lambda\Lambda^\top + \Psi$ unchanged. The latent axes are therefore not uniquely labeled. For a linear state-space model,

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad y_t = Cx_t + v_t,$$

any invertible T changes coordinates to $A' = TAT^{-1}$, $B' = TB$, $C' = CT^{-1}$, with transformed process-noise covariance, while preserving the measured trajectories. This is *similarity ambiguity*, not an estimator bug. General nonlinear latent-variable models may have even larger coordinate freedom.

8.4 Alignment and comparison of subspaces

For matrices $X, Y \in \mathbb{R}^{n\times k}$ of aligned, centered samples, the orthogonal Procrustes problem is

$$\min_{Q^\top Q = I} \|XQ - Y\|_F^2.$$

Proposition 8.2 (Procrustes solution)**.** *Let $X^\top Y = U\Sigma V^\top$ be an SVD. Then $Q_* = UV^\top$ is an optimal orthogonal alignment.*

Proof. Expanding the squared norm leaves the variable term $-2\,\mathrm{tr}(Q^\top X^\top Y)$. Writing $Q = UZV^\top$, the trace is $\mathrm{tr}(Z^\top \Sigma) \le \sum_i \sigma_i$, maximized by $Z = I$. □

Canonical correlation analysis (CCA) instead finds directions with maximal *cross-view correlation* after whitening both covariance structures. Representational similarity analysis (RSA) and centered kernel alignment (CKA) compare geometries through different invariances. They require carefully matched observations, preprocessing and independent validation sets.

Example 8.3 (Existing code)**.** `population_pca.m` produces an SVD of centered recordings. `participation_ratio.m` summarizes a covariance eigenspectrum. Neither establishes BBP significance, latent-factor uniqueness, or a well-identified inter-area communication subspace without additional modeling and tests.

Chapter 9

Dependent observations, selective inference, and causality

Time bins from one animal are typically correlated. Repeating a test at many channels, lags and frequency bands also creates selection effects. A valid mathematical result must state which dependence and selection operations it covers.

9.1 Variance of a time-series average

For a zero-mean weakly stationary time series with $\gamma(k) = \mathrm{Cov}(X_t, X_{t+k})$, direct expansion gives

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{T}\left[\gamma(0) + 2\sum_{k=1}^{T-1}\left(1 - \frac{k}{T}\right)\gamma(k)\right].$$

This *finite-sample identity* needs no central limit theorem. When $\sum_k |\gamma(k)| < \infty$, the asymptotic variance is $[\gamma(0)+2\sum_{k\geq 1}\gamma(k)]/T$. For normalized autocorrelation $\rho(k)$, an approximate effective sample size is $T_{\mathrm{eff}} = T/[1+2\sum_{k\geq 1}\rho(k)]$, provided that series is meaningful. Negative correlations can make this heuristic exceed T; it must not be blindly clipped or treated as an exact count of independent animals.

9.2 HAC and resampling under dependence

A heteroskedasticity-and-autocorrelation-consistent (HAC) variance estimator uses weighted autocovariances, for example

$$\widehat{\Omega} = \widehat{\gamma}(0) + 2\sum_{k=1}^{L}\left(1 - \frac{k}{L+1}\right)\widehat{\gamma}(k).$$

This is a Bartlett-window Newey–West form in a scalar, mean-estimation setting. Consistency requires hypotheses on stationarity, weak dependence, moments and bandwidth growth; finite samples and nonstationary trials require scrutiny. Block bootstrap preserves within-block temporal structure but is only justified under suitable dependence assumptions and a suitable block length. Randomly permuting every LFP sample is generally *not* an exchangeable null.

9.3 Multiple comparisons and post-selection

Holm's step-down procedure controls family-wise error under valid marginal p-values without an independence assumption. Benjamini–Yekutieli controls FDR under arbitrary dependence using

the harmonic correction $c_m = \sum_{j=1}^{m} 1/j$ in its standard formulation; it may be conservative. A maximum-statistic permutation test controls a family of tests if the randomization acts on the units exchangeable under the joint null. Cluster-permutation procedures use adjacency and a cluster-level statistic; rejection does not locate the exact significant onset, offset, or single electrode. After selecting the largest spectral peak and testing it on the same data, ordinary unadjusted p-values generally fail to reflect that selection. Sample splitting, independent replication or mathematically justified selective inference are possible remedies.

9.4 Observational dependence versus intervention

A directed acyclic graph (DAG) encodes a collection of conditional-independence relations under the Markov property. Paths are blocked or unblocked according to *d-separation*. If Z is an appropriate backdoor adjustment set for a proposed causal effect $X \to Y$, under causal sufficiency assumptions relative to the DAG and positivity,

$$p(y \mid \mathrm{do}(x)) = \int p(y \mid x, z) p(z)\, dz.$$

The formula fails for invalid adjustment sets, unmeasured confounding, or violated structural assumptions. In particular, a lagged correlation does not supply the DAG. Faithfulness is an additional assumption (and can fail by parameter cancellation); the common-cause principle motivates but does not uniquely identify latent confounders.

9.5 Transfer entropy and Gaussian Granger

Transfer entropy from X to Y, conditional on selected histories Y^- and Z^-, is a conditional mutual information: $TE = I(Y_t; X^- \mid Y^-, Z^-)$. For jointly Gaussian variables, with unrestricted correct linear conditional predictors and covariance determinants specified consistently,

$$TE_{X \to Y|Z} = \tfrac{1}{2} \log \frac{\det \Sigma_{\text{reduced}}}{\det \Sigma_{\text{full}}} \quad \text{(nats)}.$$

This Gaussian equivalence is established in [3]. A common multivariate Granger measure is the full log ratio; therefore it equals $2TE$ in this Gaussian setting. This is a theorem about *predictive distributions*, not an identification theorem for causal interventions. Spectral Granger additionally requires consistent VAR transfer functions or spectral factorization (e.g., Wilson methods) and model residual checks. The present `granger_causality.m` computes a temporal log-variance ratio, not spectral Granger; the existing `transfer_entropy.m` is a first-order, discrete plug-in conditional-information estimator.

Interpretation warning 9.1. The two MATLAB primitives are exploratory and do not by themselves establish causal identifiability.

Chapter 10

Oscillators, bifurcations, and stochastic dynamics

Oscillatory-looking trajectories can emerge from a dynamical system, an externally driven filter, or chance structure in a finite recording. These are mathematically distinct accounts.

10.1 Hopf normal form and an actual periodic attractor

The complex-valued ordinary differential equation $\dot{z} = (\mu + i\omega)z - |z|^2 z$ with $\omega \neq 0$ can be written $z = re^{i\theta}$, giving

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega.$$

For $\mu < 0$ the origin attracts nearby trajectories. For $\mu > 0$ the origin repels, while the circle $r = \sqrt{\mu}$ is an attracting periodic orbit. Its period is $2\pi/|\omega|$. This explicit derivation demonstrates a supercritical Hopf-type normal form; deciding whether experimental LFP oscillations arise from an actual Hopf bifurcation requires more evidence.

10.2 Phase reduction and weak coupling

A stable limit-cycle oscillator with phase coordinate θ responds to weak perturbation $u(t)$ to leading order as

$$\dot{\theta} = \omega + Z(\theta) \cdot u(t),$$

where Z is the infinitesimal phase-response curve (PRC). For two weakly coupled oscillators, averaging can produce a phase-difference equation $\dot{\psi} = \Delta\omega + \varepsilon H(\psi)$. Locking requires a stable fixed point of this equation; oscillatory phase locking need not imply anatomical synapses, and non-sinusoidal harmonics can generate apparent PAC.

10.3 jPCA and skew-symmetric dynamics

If $\dot{x} = Mx$ for a real skew-symmetric matrix $M^\top = -M$, then

$$\frac{d}{dt}\|x\|_2^2 = 2x^\top Mx = 0.$$

A real skew-symmetric matrix has orthogonal real normal form consisting of planar blocks $\left(\begin{smallmatrix} 0 & -\omega_j \\ \omega_j & 0 \end{smallmatrix}\right)$ (and possibly zeros); its nonzero eigenvalues occur as imaginary pairs $\pm i\omega_j$. Thus a fitted rotational component conserves Euclidean energy in the ideal autonomous model. The file `jpca_dynamics.m` fits such a skew generator to finite-difference derivatives[8], not a complete state-space model or a proof of physiological cycles.

10.4 Ornstein–Uhlenbeck noise and the Fokker–Planck equation

For an Itô stochastic differential equation $dX_t = a(X_t)dt + b(X_t)dW_t$, a twice differentiable function f obeys Itô's formula

$$df(X_t) = \left(af' + \tfrac{1}{2}b^2 f''\right)dt + bf'\,dW_t.$$

Its density, under regularity and boundary assumptions, evolves by the Fokker–Planck equation $\partial_t p = -\partial_x(ap) + \tfrac{1}{2}\partial_{xx}(b^2 p)$. For the Ornstein–Uhlenbeck process $dX_t = -\kappa X_t dt + \sigma dW_t$, with $\kappa > 0$, the stationary variance is $\sigma^2/(2\kappa)$ and the stationary autocovariance is $\sigma^2 e^{-\kappa|\tau|}/(2\kappa)$. An exponential autocorrelation is therefore compatible with a stochastic relaxation process and need not imply an intrinsic neuronal pacemaker or periodic oscillator.

Example 10.1 (Simulation check)**.** Simulate a damped OU fluctuation and a noisy Hopf limit cycle at the same sampling frequency. Compare autocorrelation, PSD and phase stability under different recording lengths. A narrow spectral peak alone does not establish the Hopf model.

Chapter 11

Robust estimates, information bias, and numerical stability

Analysis pipelines need both statistical robustness to unusual data points and numerical robustness to finite precision. Neither follows merely from a mathematical formula being valid.

11.1 Median and median absolute deviation

The sample median has finite-sample replacement breakdown point approaching 1/2: fewer than about half the observations cannot drive it arbitrarily far by contamination. For observations $x_1, \dots, x_T$, define MAD $= \operatorname{median}_i |x_i - \operatorname{median}_j x_j|$. For normal errors a common robust scale estimate is $1.4826\,\mathrm{MAD}$, asymptotically calibrated under the Gaussian model. For artifacted spike waveforms, median/MAD thresholding can be more stable than a mean/standard deviation, but physiological artifacts and nonstationarity can contaminate even robust estimates. An influence function formalizes the estimator's infinitesimal response to a point-mass contamination.

11.2 Empirical distribution bounds and time-rescaling

For iid observations with distribution function F, let F_T be the empirical CDF. A Dvoretzky–Kiefer–Wolfowitz bound states for $\epsilon > 0$

$$\mathbb{P}\left(\sup_x |F_T(x) - F(x)| > \epsilon\right) \le 2e^{-2T\epsilon^2}.$$

It implies uniform consistency (a Glivenko–Cantelli conclusion) in the iid setting. The KS check of rescaled interspike intervals uses an exponential or uniform null only when the conditional intensity is correctly specified and relevant transformed intervals satisfy the needed independence assumptions. Repeated fitting and testing on the same data changes calibration.

11.3 Information-theoretic chain rules and finite-sample bias

For discrete variables, the entropy chain rule is $H(X,Y) = H(X) + H(Y \mid X)$ and $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z)$. The Gaussian distribution maximizes differential entropy among random vectors with a specified positive-definite covariance, with entropy $\frac{1}{2}\log[(2\pi e)^d \det \Sigma]$. A plug-in entropy estimator from a finite discrete histogram usually has downward bias. With K effectively occupied bins, a leading Miller–Madow-style correction in *nats* is $(K-1)/(2T)$ under its asymptotic sampling assumptions. It does not cure curse-of-dimensionality or guarantee an

unbiased information estimator. Continuous estimators such as KSG have different nearest-neighbor assumptions and biases; the `mutual_information.m` implementation should not be described as using KSG without verifying code.

11.4 Conditioning, least squares, and stable algorithms

If a full-rank matrix X has spectral condition number $\kappa_2(X) = \sigma_{\max}/\sigma_{\min}$, then $\kappa_2(X^\top X) = \kappa_2(X)^2$.

Proof. The singular values of X are the square roots of the eigenvalues of $X^\top X$; hence those eigenvalues are σ_j^2 and the ratio of extremes is squared. □

This explains why explicitly solving normal equations can amplify roundoff relative to QR/SVD solutions. A backward-stable algorithm returns an exact answer for a nearby problem; it does not guarantee small forward error for ill-conditioned inputs. The FFT reduces the computation of a DFT from the naive $O(T^2)$ complex arithmetic to $O(T \log T)$ for appropriate factorizable sizes. Algorithmic efficiency does not change spectral-estimator uncertainty.

Example 11.1 (Research reproducibility)**.** When regressing spike history on population counts, report condition numbers, convergence criteria, effective independent units, model selection, bin duration and handling of omitted variables, not just a coefficient array.

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